

illness than if the wage rate were independent of pollution—clearly counterintuitive. Forestalling discussion of this point temporarily, and proceeding as in (18)–(22) above, it can be shown that

$$\frac{dC}{dP} = \frac{dS}{dP} \left[\frac{U_S}{\lambda} + w'(S)(T - L) - \frac{1}{S_D} \right] \quad (26)$$

which, making use of (18), can be expressed as

$$\frac{dC}{dP} = \frac{U_S}{\lambda} \frac{dS}{dP} - D_P - \frac{S_P}{S_D} + w'(S)(T - L) \frac{dS}{dP}. \quad (27)$$

But since $(dI^*/dP) = (-S_P/S_D)$,

$$\frac{dI^*}{dP} = \frac{dC}{dP} + D_P - \frac{U_S}{\lambda} \frac{dS}{dP} - w'(S)(T - L) \frac{dS}{dP}. \quad (28)$$

Finally, substituting (25) into (28) and rearranging, we have

$$\frac{dI^*}{dP} = [w(S) - (T - L - S)w'(S) + m] \frac{dS}{dP} + D_P - \frac{U_S}{\lambda} \frac{dS}{dP}. \quad (29)$$

Now we can resolve the apparent contradiction noted in (25). In fact, (25) is not the change in the cost of illness induced by pollution because it overlooks the fact that the pollution increase exerts a cost for every hour worked, not just the hours that one is sick. A more appropriate expression for cost of illness is the bracketed term in (29). Since $T - L - S = W$, or hours worked, this term now includes the loss of wages. The other two terms are, as before, defensive expenditures and direct disutility of illness.

(iii) Many workers lose no pay when ill because of paid sick leave. In this case, lost time due to illness still represents a cost to the economy, but one borne by employers. One can therefore argue that paid sick leave does not affect the total economic benefits of pollution control, but only their distribution.

This argument assumes implicitly that the availability of paid sick leave will not affect the efforts of individuals to avoid sickness.⁹ To drop this assumption, we must also drop the full-income constraint, which permits leisure and income to be traded off without restriction. Instead, suppose an individual receives a wage w per hour for W hours of work per year, regardless of whether those hours are actually worked. Let $Y = wW$ denote income, and suppose further that the distribution of work hours is fixed, as

⁹ This conceptual problem should be kept distinct from the empirical problem of estimating morbidity when workers are covered by paid sick leave. What is usually observed in morbidity studies is not sickness itself but days of illness reported. Sick leave obviously can affect whether an individual responds to illness by staying home from work.

in the 9-to-5 workday. In the previous model illness affects utility directly, by taking up time that could be devoted to labor or leisure, and by inducing out-of-pocket costs. With paid sick leave, the illness that occurs during working hours affects utility and medical costs but the individual's money income is not affected. If we assume that time spent ill is as likely to occur during working as during leisure hours, we can put the proportion q of illness occurring during leisure at $q = (T - W)/T$, the proportion of nonwork hours.

In this model the maximum number of hours that can be worked per year is W . The actual time spent working is $W - (1 - q)S$, the nominal work time less sick leave taken. The individual is no longer able to trade work for leisure without restriction, and the marginal value of leisure is no longer equal to the wage rate. The problem now is to maximize utility subject to both a money and a time constraint. The Lagrangian is

$$\mathcal{L} = U(X, L, S) + \lambda(Y - D - X - mS) + \gamma(T - W - L - qS). \quad (30)$$

The marginal change in income Y necessary to keep utility constant while pollution changes is again

$$\frac{dY}{dP} = -\frac{V_P}{V_Y} = -\frac{S_P}{S_D}. \quad (31)$$

In this case the employer also benefits from improvements in air quality, due to a reduction in sick-leave payments. Assuming that the wage rate equals the value of a day's output by a worker,¹⁰ we can represent the benefit to the employer of a marginal reduction in pollution as

$$(1 - q)w \frac{dS}{dP}. \quad (32)$$

Combining (31) and (32), the social benefit of a reduction in pollution, for each worker in the population, is

$$\text{benefit} = (1 - q)w \frac{dS}{dP} - \frac{S_P}{S_D}. \quad (33)$$

Above, we showed that the costs of illness plus reductions in defensive measures underestimate the true benefits of pollution control. Once sick leave is introduced, this may no longer be the case. From the first-order conditions for maximizing (30), we have

$$m = \frac{U_S}{\lambda} - \frac{1}{S_D} - \frac{\gamma}{\lambda}q. \quad (34)$$

¹⁰If the employer expects a certain number of sick days each year and sets wages accordingly, the value of a day's output might exceed the wage rate slightly.

The benefit estimate given by the cost-of-illness approach is

$$\begin{aligned} \text{COI} &= [w + m] \frac{dS}{dP} \\ &= w \frac{dS}{dP} + \frac{U_S}{\lambda} \frac{dS}{dP} - \frac{\gamma}{\lambda} q \frac{dS}{dP} - \frac{1}{S_D} [S_D D_P + S_P]. \end{aligned} \quad (35)$$

Total benefits B equal the sum of those occurring to the worker and his employer, and in this case equal

$$B = \text{cost of illness} + D_P - \frac{U_S}{\lambda} \frac{dS}{dP} + \left(\frac{\gamma}{\lambda} - w \right) q \frac{dS}{dP}. \quad (36)$$

Therefore, the benefit associated with a change in P equals the cost of illness plus three additional terms. The first two, representing the change in defensive expenditures and the direct disutility of illness, have been encountered before. The third term is a correction that takes account of the fact that sickness outside of working hours is no longer valued at the wage rate (the ratio γ/λ is the individual's marginal value of leisure time). If leisure has a value exceeding the wage rate, this term is positive and we can again conclude that the cost-of-illness plus defensive expenditures underestimate benefits. Even if leisure time is given a zero value, defensive expenditures plus the cost of illness adjusted by the time that the individual actually spends at work provides an underestimate. In fact, most COI studies implicitly give leisure a zero value by counting only the duration of missed work or by valuing a day of illness at the daily wage, even though individuals work only part of the day.

4. CONCLUSION

Our principal objective in this paper has been to compare ad hoc approaches to health benefit estimation with an expression of willingness to pay derived from a model of individual utility maximization. The most prominent of the ad hoc approaches is what we have called the cost of illness, which sums out-of-pocket health care expenses and lost wages. Another is the notion of defensive expenditures, discussed directly by Fisher and Zeckhauser [5], Courant and Porter [3] and, in another context, Shibata and Winrich [12]. We have shown that the cost-of-illness plus defensive expenditures normally underestimate "true" benefits. We found it was possible for defensive expenditures alone to overestimate the willingness to pay to avoid increased pollution, but only if such expenditures improve health in spite of the pollution increase.

Our finding proved robust with respect to several variations in our basic model. True benefits exceed those derived from summing defensive expenditures and the cost of illness when wages were assumed to be affected

by pollution-induced illness, and when pollution was assumed to enter the utility function both directly as well as indirectly via its effect on sickness and, hence, the labor-leisure tradeoff. Only when our basic model was extended to include the case of paid sick leave was there indeterminacy about the relationship between the traditional cost-of-illness plus defensive expenditures and willingness to pay as derived from the model. Even in this case we found an underestimate when valuation of lost wages was limited, as is customary, to time actually spent on the job.

A final observation concerns the estimability of the conceptually correct benefit measure. The expression that we derived for willingness to pay, $-S_P/S_D$, is observable in principle. Bringing this term to life, so to speak, is more difficult than may at first appear. It requires detailed information on both the market and the nonmarket measures that individuals take to defend themselves against pollution and other environmental threats. Until such data are available, our conclusions suggest that the cost-of-illness approach can be used as a lower bound for true benefits in most interesting cases.

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